

Prop complemento ortogonal subespacio cerrado

Proposición 1. Sea H un espacio prehilbert y $A \subset H$ un subconjunto

$$\implies A^\perp \text{ es un subespacio cerrado de } H.$$

Demostración: Sean $x, y \in A^\perp$ y $\alpha, \beta \in \mathbb{K}$. Entonces,

$$\forall a \in A : \langle \alpha x + \beta y, a \rangle = \alpha \langle x, a \rangle + \beta \langle y, a \rangle = 0,$$

luego $\alpha x + \beta y \in A^\perp$. Es decir, $A^\perp$ es un subespacio de H .

Sea ahora $(x_n)_{n \in \mathbb{N}} \subset A^\perp$ tal que $\lim_n x_n = x \in H$. Por continuidad del producto interno,

$$\forall a \in A : \langle x, a \rangle = \left\langle \lim_{n \rightarrow \infty} x_n, a \right\rangle = \lim_{n \rightarrow \infty} \langle x_n, a \rangle = 0 \implies x \in A^\perp. \quad \blacksquare$$

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